Opinion Formation on Networks: The DeGroot Model and Its Extensions

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Abstract—In this paper, we take a closer look at the DeGroot model (1974) and analyze its convergence and network properties. We show that if the weight matrix $\boldsymbol{A}$ is primitive, then the model converges to a consensus. We also prove that this condition can be verified by checking whether the corresponding graph is strongly connected and aperiodic. Moreover, we show that the connectedness of the graph provides an upper and lower bound for the second largest eigenvalue of $\boldsymbol{A}$, which in turn gives us an upper bound for the convergence speed of the model. Finally, we take a look at the Friedkin-Johnsen model (1990, 1999) and the Hegselmann-Krause model (2002). For the former, we show that the model converges if the product $\boldsymbol{DA}$ of the *susceptibility* matrix $\boldsymbol{D}$ and $\boldsymbol{A}$ is irreducible. For the latter, we provide a condition that ensures the convergence of the model. To explain the concepts, we provide some independent proofs and observations.

1 Introduction

As individuals interact and exchange views, local patterns of agreement or disagreement can propagate through social networks, leading to large-scale outcomes such as consensus, polarization, or fragmentation. The question of how, for example, echo chambers form and influence opinions is a strongly debated topic in our society.

In order to better understand these phenomena, many attempts have been made to analyze and model the formation of opinions, like the voter model [1], Ising model [2], Friedkin and Johnsen model [3] and Hegselmann-Krause model [4]. Another well-known example is the DeGroot model [5] which presents a rather simple, iterative process (see in Section 3). Given an instance of this model, we can then analyze or calculate properties such as whether the group will reach a consensus, what consensus they reach, and how fast they will reach it (see in Section 3.1). Moreover, the model can be represented and visualized as a graph with the nodes and edges representing who is communicating with whom. Thereby, we can use features of these graphs as conditions and explanations for the aforementioned properties (see in Section 3.2).

As the DeGroot model is a rather simple abstraction of opinion formation in real social networks, it does not capture all aspects of social interaction. Therefore, we will take a look at two extensions of the DeGroot Model (see in Section 4). First, the Friedkin-Johnsen model and second, the Hegselmann-Krause model.

2 Preliminaries and Notation

In this paper, we use $\mathbf{1}$ to denote the vector with all-ones and $\boldsymbol{I}$ to denote the identity matrix. Similarly, the vector with all-zeros is referred to as $\mathbf{0}$. For a given vector, $(.)^\mathsf{T}$ denotes the transpose of the vector. We use the following style for vectors $\boldsymbol{p}$, matrices $\boldsymbol{M}$, and graphs $\mathcal{G}$. Moreover, $V(\mathcal{G})$ denotes the set of vertices and $E(\mathcal{G})$ the set of edges in $\mathcal{G}$. For a matrix $\boldsymbol{M}$, the notation $(\boldsymbol{M})_{ij}$ refers to the entry in the i-th row and the j-th column of $\boldsymbol{M}$. Furthermore, we use the notation $\underline{n} = \{1, \dots, n\}$.

A row-stochastic matrix $\boldsymbol{A} \in [0,1]^{n\times n}$ is defined such that all its entries are non-negative and its row vectors sum up to 1, i.e., $\sum_{j=1}^{n} a_{ij} = 1$ for all $i \in \underline{n}$. A row-substochastic matrix $\boldsymbol{A} \in [0,1]^{n\times n}$ is a non-negative matrix for which all row vectors sum up to at most 1 and at least one row vector sums up to less than 1, i.e., $\sum_{j=1}^{n} a_{ij} \leq 1$ for all $i \in \underline{n}$ and $\sum_{j=1}^{n} a_{ij} < 1$ for at least one $i \in \underline{n}$. Furthermore, a digraph describes a directed graph, meaning every edge has a direction. A graph is strongly connected if and only if every node $v \in V(\mathcal{G})$ is reachable from every other node $u \in V(\mathcal{G})$ (with $v \neq u$), i.e., there exists a directed path from every node u to v. A sink $\mathcal{S}$ is a strongly connected subgraph, i.e., $V(\mathcal{S}) \subset V(\mathcal{G})$ and $E(\mathcal{S}) \subset E(\mathcal{G})$, that has no node $i \in V(\mathcal{S})$, such that there exists a node $j \notin V(\mathcal{S})$ with $(i,j) \in E(\mathcal{G})$. Lastly, a graph is aperiodic if and only if there exists no $k \in \mathbb{N}_{>1}$, that divides the length of every possible path from any node v back to itself in the graph.

3 DeGroot model

The DeGroot model considers a group of n individuals who discuss their own opinions on a given parameter θ, that describes the subject, with one another. We will introduce this model, as done in [5]. Each individual has their own opinion value for this unknown parameter θ, which we will refer to as v_i for $i \in \underline{n}$. The model uses the idea that interaction with other people in the group can cause an individual i to change their opinion value v_i based on new information or expertise provided by other individuals.

The model assumes that this revision of the individual's opinion is a linear combination of the opinion values $v_1, \dots, v_n$. To model this revision, we therefore define a_{ij} which denotes the weight that individual i assigns to the opinion of individual j. If, for example, individual j is a leading expert on the value of θ, then individual i might want to assign a greater weight to the opinion of individual

j. Similarly, individual i can also assign a large weight to their own opinion and a small total weight to the opinions of other individuals. The weights are assigned by the individuals without any knowledge of the other individuals' assigned weights.

Let v_{ik} denote the opinion value of individual i after revision k, then the revised opinion value of the individual i is calculated as

$$v_{i(k+1)} = \sum_{j=1}^{n} a_{ij} v_{jk} \ . \tag{1}$$

It is assumed that $a_{ij} \geq 0$ and $\sum_{j=1}^{n} a_{ij} = 1$, meaning that the weights assigned by each individual sum up to 1. Therefore, the matrix representing the weights

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix},$$

is row-stochastic, see in Section 2.

If we let the vector $\boldsymbol{p_0} = (v_1, \dots, v_n)^\mathsf{T}$ denote the initial opinion values and $\boldsymbol{p_k} = (v_{1k}, \dots, v_{nk})^\mathsf{T}$ denote the opinion values after k revisions, we can express a given revision as

$$\boldsymbol{p_k} = \boldsymbol{A}\boldsymbol{p_{k-1}} = \boldsymbol{A}^k \boldsymbol{p_0} \ , \tag{2}$$

for $k = 1, 2, \dots$. This process of revising the opinions is then repeated until the vector $\boldsymbol{p_k}$ converges asymptotically, with

$$\boldsymbol{p_k} = \boldsymbol{p_{k+1}} = \boldsymbol{A}\boldsymbol{p_k} \ . \tag{3}$$

As $\boldsymbol{p_k} = \boldsymbol{A}^k \boldsymbol{p_0}$, see Eq. (2), it follows for Eq. (3) that

$$\boldsymbol{A}^k \boldsymbol{p_0} = \boldsymbol{A}^k \boldsymbol{A} \boldsymbol{p_0} \ , \tag{4}$$

i.e., $\lim_{k\to\infty} \boldsymbol{A}^k$ converges. We will refer to this converged matrix as $\boldsymbol{A^{fin}}$ and to the converged final opinion values as $\boldsymbol{p^{fin}} = (v_1^{fin}, \dots, v_n^{fin})^\mathsf{T}$.

3.1 Convergence Properties

When the opinion values have converged asymptotically, as seen in Eq. (3), we can stop the iterative revision process. If the values of all n components in $\boldsymbol{p^{fin}}$ are identical, with $v_1^{fin} = \dots = v_n^{fin}$, then the n individuals of the group have reached an identical opinion value. In this case, we say that the group has reached a consensus.

With this definition of a consensus, we can now answer four central questions in the following subsections:

1) When does the model converge? (Section 3.1.1)
2) When is consensus reached? (Section 3.1.2)
3) What consensus is reached? (Section 3.1.3)
4) How can we measure the speed of convergence? (Section 3.1.4)

3.1.1 Convergence of the DeGroot Model

To determine when a consensus is reached, we first have to ensure that the model converges. As seen in Eq. (4), this is true if $\lim_{k\to\infty} \boldsymbol{A}^k$ exists. However, this does not always have to be the case, as can be seen in the following example.

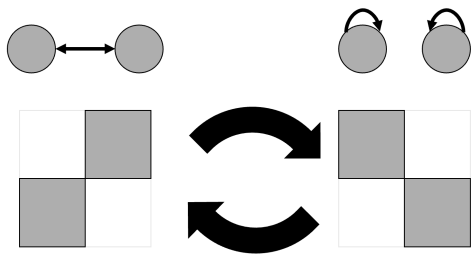


Fig. 1: Top: The given graph. Bottom: A heatmap of the corresponding adjacency matrix (gray: $a_{ij} = 1$, white: $a_{ij} = 0$)

If the row-stochastic matrix $\boldsymbol{M}$, shown on the left in Fig. 1, is multiplied by itself, it takes the form shown on the left for uneven powers $\boldsymbol{M}^{2k+1}$, and the form on the right for even powers $\boldsymbol{M}^{2k}$. Therefore, $\lim_{k\to\infty} \boldsymbol{M}^k$ does not exist. The real-world counterpart for this example would be a pair of individuals who swap their two opinions at each revision.

Thus, we need to find necessary conditions for the convergence of row-stochastic matrices. Every square matrix $\boldsymbol{A} \in \mathbb{C}^{n\times n}$ is similar to a block diagonal matrix $\boldsymbol{J} \in \mathbb{C}^{n\times n}$, also called the Jordan normal form of $\boldsymbol{A}$, meaning $\boldsymbol{A} = \boldsymbol{T}\boldsymbol{J}\boldsymbol{T}^{-1}$ for an invertible matrix $\boldsymbol{T} \in \mathbb{C}^{n\times n}$. The blocks on the diagonal of the Jordan matrix are called Jordan blocks $\boldsymbol{J_i}$ and have the form

$$\boldsymbol{J_i} = \begin{bmatrix} \lambda_i & 1 & 0 & \dots & 0 \\ 0 & \lambda_i & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 & 0 \\ 0 & \ddots & 0 & \lambda_i & 1 \\ 0 & 0 & \dots & 0 & \lambda_i \end{bmatrix},$$

where λ_i is the i-th eigenvalue of $\boldsymbol{A}$.

As $\boldsymbol{A}^k = \boldsymbol{T}\boldsymbol{J}^k\boldsymbol{T}^{-1}$, we can use this equation to find the limit of $\boldsymbol{A}^k$, as $\lim_{k\to\infty} \boldsymbol{A}^k = \boldsymbol{T}(\lim_{k\to\infty} \boldsymbol{J}^k)\boldsymbol{T}^{-1}$. Therefore, it suffices to show that $\boldsymbol{J}^k$ converges, to prove that $\boldsymbol{A}^k$ converges.

As the k-th power of a Jordan block $\boldsymbol{J_i} \in \mathbb{C}^{l\times l}$ with the eigenvalue λ_i has the form

$$\boldsymbol{J_i^k} = \begin{bmatrix} \lambda_i^k & \binom{k}{1}\lambda_i^{k-1} & \binom{k}{2}\lambda_i^{k-2} & \dots & \binom{k}{l-1}\lambda_i^{k-l+1} \\ 0 & \lambda_i^k & \binom{k}{1}\lambda_i^{k-1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \binom{k}{2}\lambda_i^{k-2} \\ 0 & \ddots & 0 & \lambda_i^k & \binom{k}{1}\lambda_i^{k-1} \\ 0 & 0 & \dots & 0 & \lambda_i^k \end{bmatrix},$$

we can conclude the following conditions, shown in [6], for the convergence of matrix $\boldsymbol{A}$. First, the magnitudes of all of its eigenvalues λ_i must be $|\lambda_i| \leq 1$. If $|\lambda_i| > 1$, then all non-zero entries $(\boldsymbol{J_i})_{ij}$ of $\boldsymbol{J_i}$ diverge with $\lim_{k\to\infty}(\boldsymbol{J_i^k})_{ij} = \infty$. When some Jordan block $\boldsymbol{J_i}$ diverges for $\lim_{k\to\infty}$, then trivially the Jordan matrix $\boldsymbol{J}$, and thus also $\boldsymbol{A}$, diverge as well. Second, if $\lambda_j = -1$ is an eigenvalue of $\boldsymbol{A}$, then the entries λ_j on the diagonal of the corresponding Jordan block $\boldsymbol{J_j}$ diverge for $\lim_{k\to\infty} \boldsymbol{J_j^k}$, as the powers λ_j^k oscillate between -1 and 1. Thus, $\boldsymbol{A}$ also oscillates, as can be observed in Fig. 1. Third, if $\lambda_m = 1$ is an eigenvalue of $\boldsymbol{A}$, it needs to be semisimple, meaning that its geometric

and algebraic multiplicities are equal. In other words, the corresponding Jordan blocks $\boldsymbol{J_m}$ must have size 1, i.e., $\boldsymbol{J_m} = [\lambda_m]$. If this condition is not satisfied, the entries $(\boldsymbol{J_m})_{ij}$ with $j > i$ diverge for $\lim_{k\to\infty}(\boldsymbol{J_m^k})_{ij} = \infty$. In the following, we will refer to the eigenvalue 1 of $\boldsymbol{A}$, also called the dominant eigenvalue, as λ_1.

Additionally, because $\boldsymbol{T}$ is invertible, we know that $\boldsymbol{T}, \boldsymbol{T}^{-1} \neq \boldsymbol{0_{n\times n}}$. Therefore, if $\boldsymbol{A}$ has the semisimple eigenvalue λ_1 and the limit of $\boldsymbol{A}$ exists, then

$$\lim_{k\to\infty} \boldsymbol{A}^k = \boldsymbol{T}(\lim_{k\to\infty} \boldsymbol{J}^k)\boldsymbol{T}^{-1} \neq \boldsymbol{0_{n\times n}} \ ,$$

as the Jordan blocks corresponding to λ_1 converge against $\lim_{k\to\infty} \boldsymbol{J_1^k} = \lim_{k\to\infty} [1]^k = [1]$, i.e., $\lim_{k\to\infty} \boldsymbol{J}^k \neq \boldsymbol{0_{n\times n}}$. We refer to such a matrix $\boldsymbol{A}$ as semi-convergent.

Because $\boldsymbol{A}$ is row-stochastic, by definition

$$\boldsymbol{A1} = \boldsymbol{1} = 1 \cdot \boldsymbol{1}, \text{with } \boldsymbol{1} \neq \boldsymbol{0} \ , \tag{5}$$

as all rows have unit sum. Thus, λ_1 is an eigenvalue of all matrices $\boldsymbol{A}$ of the model, which are consequently semi-convergent, if a limit exists.

We can summarize these findings in the following Theorem 3.1, found in [6].

Theorem 3.1. *A row-stochastic matrix $\boldsymbol{A}$ is semi-convergent (i.e., $\lim_{k\to\infty} \boldsymbol{A}^k \neq \boldsymbol{0_{n\times n}}$ exists) if and only if:*

(i) 1 is an eigenvalue
(ii) 1 a semisimple eigenvalue, and
(iii) all other eigenvalues have magnitude less than 1.

As shown, condition (i) of Theorem 3.1 holds for all row-stochastic matrices. We will discuss under which circumstances the other conditions hold in the following section.

3.1.2 Reaching a Consensus

To now determine when a consensus is reached, we will utilize the Perron-Frobenius Theorem 3.2, namely, irreducible and primitive matrices.

Definition 3.1 (Irreducible and primitive matrices)**.** *For $n \geq 2$, a non-negative matrix $\boldsymbol{A} \in \mathbb{R}_{\geq 0}^{n\times n}$ is*

(i) irreducible if $\sum_{k=0}^{n-1} \boldsymbol{A}^k > 0$
(ii) primitive if there exists a $k \in \mathbb{N}$, such that $\boldsymbol{A}^k > 0$.

Moreover, a primitive matrix is necessarily also irreducible, see [6].

When applying the Perron-Frobenius Theorem to row-stochastic matrices $\boldsymbol{A}$, we get the following, as found in [6].

Theorem 3.2 (Perron-Frobenius Theorem)**.** *Let the row-stochastic $\boldsymbol{A} \in [0,1]^{n\times n}$ with $n \geq 2$, then if $\boldsymbol{A}$ is irreducible,*

(i) the eigenvalue λ_1 is simple.
If additionally $\boldsymbol{A}$ is primitive, then
(ii) the eigenvalue λ_1 satisfies $\lambda_1 > |\lambda_i|$ for all other eigenvalues λ_i of $\boldsymbol{A}$.

Therefore, according to the Perron-Frobenius Theorem 3.2, if $\boldsymbol{A}$ is irreducible and row-stochastic, then λ_1 is simple, which is a sufficient condition for semisimple, thus satisfying condition (ii) of Theorem 3.1. Moreover, if additionally $\boldsymbol{A}$ is primitive, then all eigenvalues $\lambda_i \neq 1$ have magnitude less than 1, i.e., $\lambda_1 > |\lambda_i|$, thus satisfying condition (iii) of Theorem 3.1. It can therefore be concluded that if a row-stochastic matrix is primitive, then $\boldsymbol{A}$ is semi-convergent, i.e., the model converges.

However, it can often be computationally expensive to check whether a given matrix is irreducible or primitive. Therefore, we will now relate these conditions to graph properties, which are typically easier to verify.

We can translate any instance of the DeGroot model into a digraph $\mathcal{G} = (V, E)$ with the nodes $V(\mathcal{G}) = \{1, \dots, n\}$ representing the individuals and the edges $E(\mathcal{G}) = \{(i,j) | a_{ij} > 0\}$ with the corresponding weighted adjacency matrix $\boldsymbol{A}$, defined in Section 3. We can now observe that a path of length k exists from node i to node j in $\mathcal{G}$ if and only if for the k-th power of the adjacency matrix, $(\boldsymbol{A}^k)_{ij} > 0$. For this, we offer an independent proof.

Proof. Let $\boldsymbol{A} \in [0,1]^{n\times n}$. There trivially exists a path of length 1 from node i to node j, precisely when $(\boldsymbol{A})_{ij} = a_{ij} > 0$, i.e., there exists an edge from i to j. Assume that the observation is correct for any $k \in \underline{n}$. If there exists a path of length $k+1$ from i to j, then there exists a path of length k from i to some node $l \in \underline{n}$ with $(l,j) \in E(\mathcal{G})$, i.e., $(\boldsymbol{A})_{lj} > 0$. As $(\boldsymbol{A}^{k+1})_{ij} = (\boldsymbol{A}^k \cdot \boldsymbol{A})_{ij} = \sum_{m=1}^{n} (\boldsymbol{A}^k)_{im} \cdot (\boldsymbol{A})_{mj} \geq (\boldsymbol{A}^k)_{il} \cdot (\boldsymbol{A})_{lj} > 0$, because due to the induction hypothesis $(\boldsymbol{A}^k)_{il} > 0$. In the other direction, if $(\boldsymbol{A}^{k+1})_{ij} > 0$, then there must be at least one $l \in \underline{n}$, such that $(\boldsymbol{A}^k)_{il} \cdot (\boldsymbol{A})_{lj} > 0$. Thus, using the induction hypothesis, there must exist a path of length k from i to l and $(l,j) \in E(\mathcal{G})$. Therefore, there exists a path of length $k+1$ from i to j over l. □

This means that our digraph $\mathcal{G}$ is strongly connected, if and only if for every pair of nodes (i,j) with $i \neq j$, $(\boldsymbol{A}^k)_{ij} > 0$, for a $k < n$. It trivially suffices to ensure this property for $k < n$, as if a path of length $k \geq n$ exists, then at least $n+1$ nodes are visited, i.e., at least one node is visited multiple times. Consequently, a loop exists in this path which can be removed, resulting in a path of length $k < n$. Thus, $\mathcal{G}$ is strongly connected if and only if $\sum_{k=0}^{n-1} \boldsymbol{A}^k > 0$, i.e., $\boldsymbol{A}$ is irreducible, see Definition 3.1. Moreover, as shown in [6], the weighted adjacency matrix $\boldsymbol{A}$ is primitive if and only if $\mathcal{G}$ is strongly connected and aperiodic. Thus, these two properties ensure the asymptotic convergence of the model.

Now, we can examine when the model reaches consensus. As stated above, all eigenvalues of a row-stochastic matrix $\boldsymbol{A}$ must have a magnitude of less than 1, except for the eigenvalue λ_1, for $\boldsymbol{A}^k$ to converge. Moreover, the eigenvalue λ_1 needs to be semisimple, for the corresponding Jordan blocks to converge. Therefore, if we let $m_g \in \underline{n}$ denote the geometric multiplicity (and thus also the algebraic multiplicity) of eigenvalue λ_1, then the limit of the Jordan matrix $\lim_{k\to\infty} \boldsymbol{J}^k$ is a diagonal matrix with a 1 in m_g entries of the diagonal. Let $\boldsymbol{c_1}, \dots, \boldsymbol{c_{m_g}}$ denote the columns of $\boldsymbol{T}$ and $\boldsymbol{r_1^\mathsf{T}}, \dots, \boldsymbol{r_{m_g}^\mathsf{T}}$ denote the rows of $\boldsymbol{T}^{-1}$ that correspond to the Jordan blocks for the eigenvalue λ_1. Moreover, let $(c_i)_j$ and $(r_i)_j$ denote the jth entry of these vectors. Then

$$\begin{aligned}\boldsymbol{T}(\lim_{k\to\infty} \boldsymbol{J}^k)\boldsymbol{T}^{-1} &= \begin{bmatrix}\boldsymbol{c_1} & \dots & \boldsymbol{c_{m_g}}\end{bmatrix} \cdot \begin{bmatrix}\boldsymbol{r_1} & \dots & \boldsymbol{r_{m_g}}\end{bmatrix}^\mathsf{T} \\ &= \begin{bmatrix}(c_1)_1 & \dots & (c_{m_g})_1 \\ \vdots & \vdots & \vdots \\ (c_1)_n & \dots & (c_{m_g})_n\end{bmatrix} \cdot \begin{bmatrix}(r_1)_1 & \dots & (r_1)_n \\ \vdots & \vdots & \vdots \\ (r_{m_g})_1 & \dots & (r_{m_g})_n\end{bmatrix}\end{aligned} \tag{6}$$

To reach an asymptotic consensus, all rows of $\boldsymbol{A^{fin}}$ must be equal. Thus, all columns $\boldsymbol{c_1}, \ldots, \boldsymbol{c_{m_g}}$ of $\boldsymbol{T}$ need to be equal, see Eq. (6).

As, $\boldsymbol{A} = \boldsymbol{TJT^{-1}} \iff \boldsymbol{AT} = \boldsymbol{TJ} \iff \boldsymbol{T^{-1}A} = \boldsymbol{JT^{-1}}$, if λ_1 is semisimple, then $\boldsymbol{Ac_i} = \lambda_1 \boldsymbol{c_i}$ and $\boldsymbol{r_i A} = \lambda_1 \boldsymbol{r_i}$, i.e., $\boldsymbol{c_i}$ is the right eigenvector and $\boldsymbol{r_i^\mathsf{T}}$ the left eigenvector of λ_1, as shown in [6]. Because the number of dimensions of an eigenspace is equal to the geometric multiplicity of the corresponding eigenvalue, the eigenspace of λ_1 has m_g dimensions [7]. Thus, all of the m_g eigenvectors c_i of λ_1 need to be linearly independent. Therefore, they cannot be equal if the geometric multiplicity $m_g > 1$, meaning the rows of $\boldsymbol{A^{fin}}$ are equal if and only if $m_g = 1$, i.e., λ_1 is simple.

Thus, the model reaches an asymptotic consensus if and only if the row-stochastic matrix $\boldsymbol{A}$ is semi-convergent and λ_1 is simple. Therefore, according to the Perron-Frobenius Theorem 3.2, the model converges when $\boldsymbol{A}$ is primitive. However, $\boldsymbol{A}$ being primitive is not a necessary condition to reach an asymptotic consensus. Intuitively, as all final opinion values depend on the individual i corresponding to the simple eigenvalue λ_1, all other individuals must be connected to this individual i. Otherwise, their opinion value could not be influenced by individual i. Thus, the node corresponding to individual i needs to be globally reachable, i.e., reachable from all other nodes. In other words, there must exist a globally reachable node, which corresponds precisely to λ_1, for the model to asymptotically reach a consensus. This condition is sufficient to determine whether asymptotic consensus is reached, see [6].

3.1.3 Calculating the Consensus

However, we are interested not only in when a consensus is reached, but also in what consensus is reached. Therefore, we will take a look at how the consensus can be calculated in case it is reached.

As we concluded in Section 3.1.2, the model converges to a consensus if and only if $\boldsymbol{A}$ is semi-convergent and λ_1 is simple. Thus, using Eq. (6), $\lim_{k\to\infty} \boldsymbol{A}^k = \boldsymbol{c} \cdot \boldsymbol{r}$, where $\boldsymbol{c} = [c_1, \ldots, c_n]^\mathsf{T}$ and $\boldsymbol{r} = [r_1, \ldots, r_n]$ are the right and left eigenvectors of λ_1.

Because $\mathbf{1}$ is an eigenvector of every row-stochastic matrix, as shown in Eq. (5), the right eigenvector $\boldsymbol{c}$ can be chosen as $\boldsymbol{c} = \mathbf{1}$. Then

$$\lim_{k\to\infty} \boldsymbol{A}^k = \mathbf{1}\boldsymbol{r} = \begin{bmatrix} \boldsymbol{r} \\ \vdots \\ \boldsymbol{r} \end{bmatrix}_{n\times n} = \begin{bmatrix} r_1 & \ldots & r_n \\ \vdots & \vdots & \vdots \\ r_1 & \ldots & r_n \end{bmatrix} . \tag{7}$$

To calculate the consensus, we now need to combine Eq. (7) with Eq. (2), such that

$$\boldsymbol{p^{fin}} = \lim_{k\to\infty} \boldsymbol{p}^k = (\lim_{k\to\infty} \boldsymbol{A}^k)\boldsymbol{p_0} = \begin{bmatrix} \boldsymbol{r} \\ \vdots \\ \boldsymbol{r} \end{bmatrix}_{n\times n} \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} . \tag{8}$$

Therefore, the consensus is equal to

$$v_i^{fin} = r_1 v_1 + r_2 v_2 + \cdots + r_n v_n \ ,$$

for all $i \in \underline{n}$.

In summary, only the left eigenvector $\boldsymbol{r}$ of λ_1 of the matrix $\boldsymbol{A}$ is required, in addition to the vector of the initial opinion values $\boldsymbol{p_0}$, to calculate the final opinion values.

Moreover, we can interpret the structure of this left eigenvector $\boldsymbol{r}^\mathsf{T}$ as a weighted average over the initial opinion values in $\boldsymbol{p_0}$. The weights intuitively depend on the centrality of the corresponding node, i.e., how many other nodes are connected to it. Thus, if, for example, a given instance of the model is not strongly connected but has one sink component (see Section 2), then only the weights corresponding to nodes in the sink are non-zero.

3.1.4 Measuring the Speed of Convergence

Finally, after analyzing when and what consensus is reached, we will examine whether we can measure the speed of convergence, i.e., how long it takes for the group to asymptotically reach a consensus.

Rather intuitively, we can observe that the speed of convergence depends on the eigenvalues of matrix $\boldsymbol{A}$. As we have shown in Section 3.1.1, the convergence of A can be represented by its Jordan matrix, as $\boldsymbol{A}^k = \boldsymbol{TJ}^k\boldsymbol{T}^{-1}$. The matrix $\boldsymbol{J}^k$ converges when all eigenvalues $|\lambda_i| < 1$, i.e., all eigenvalues except λ_1, have converged against 0, with $\lim_{k\to\infty} \lambda_i^k = 0$. Therefore, the speed of convergence depends on the size of the eigenvalues of $\boldsymbol{A}$. More specifically, the convergence rate is dependent on the largest eigenvalue λ_{max}, with $|\lambda_{max}| \geq |\lambda_i|$ for all $i \in \{2, \ldots, n\}$.

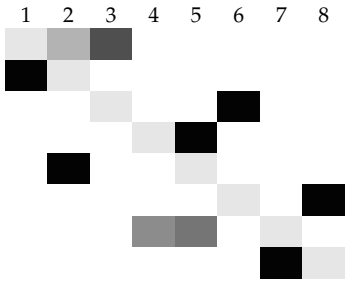
For this λ_{max}, it trivially holds that $\lambda_{max}^k \geq \lambda_i^k$, with $k > 0$, for all $i \in \{2, \ldots, n\}$. Therefore, we can define the upper bound

$$|v_{ik} - \boldsymbol{r}\boldsymbol{p_0}| = |v_{ik} - v_i^{fin}| = O(\lambda_{max}^k) \ ,$$

as in [8]. This bound describes that the distance between all opinion values in revision k and the final opinion value depends on the size of the eigenvalue λ_{max}. Thus, we know for all pairs of semi-convergent, row-stochastic matrices $\boldsymbol{A}$ and $\boldsymbol{A'}$, that if $\lambda_{max} < \lambda'_{max}$, then $\lim_{k\to\infty} \boldsymbol{A}^k$ converges faster asymptotically than $\lim_{k\to\infty} (\boldsymbol{A'})^k$. This can be observed in Fig. 2 and Fig. 3. The eigenvalue λ_{max} of the adjacency matrix used in Fig. 3 is much smaller in magnitude than that of the adjacency matrix used in Fig. 2. Thus, the model in Fig. 3 converges faster asymptotically, than the model in Fig. 2, as can be seen in the figures.

Therefore, we can measure the convergence rate of an instance of the model by calculating the difference between $\lambda_1 - |\lambda_{max}|$. This eigengap between the dominant eigenvalue λ_1 and λ_{max} is also referred to as the spectral gap.

TABLE 1: Adjacency matrix used in Fig. 2 with $|\lambda_{max}| \approx 0.932$. A darker square corresponds to an entry with a larger relative value.



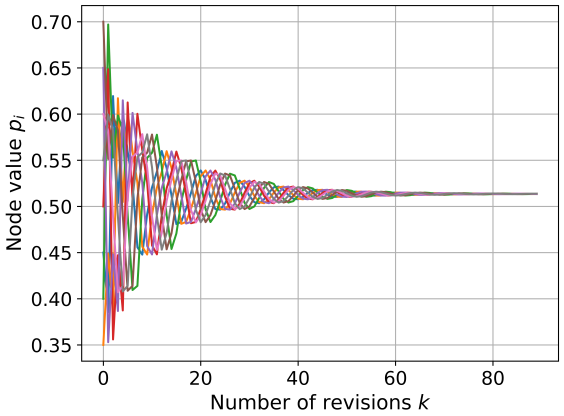
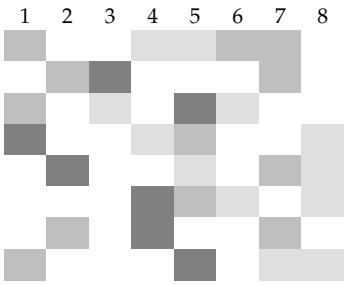


Fig. 2: The convergence towards a consensus for an instance of the DeGroot model with the adjacency matrix shown in Table 1.

TABLE 2: Adjacency matrix used in Fig. 3 with $|\lambda_{max}| \approx 0.449$. A darker square corresponds to an entry with a larger relative value.



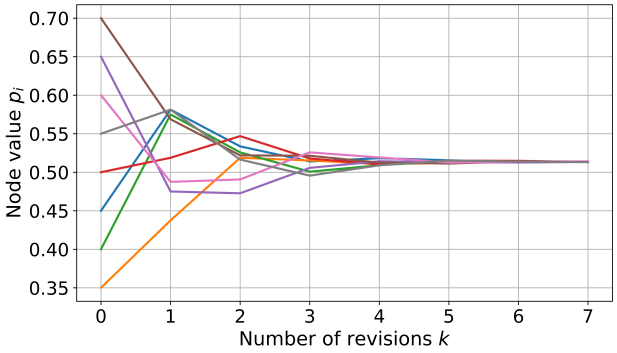


Fig. 3: The convergence towards a consensus for an instance of the DeGroot model with the adjacency matrix shown in Table 2.

3.2 Influence of Network Properties on the Convergence

As shown in Section 3.1.2, we can represent the DeGroot model as a digraph $\mathcal{G}$ with the individuals as nodes and $\boldsymbol{A}$ as the weighted adjacency matrix. Therefore, we will look at how certain network and graph properties can influence the model.

First, we will look at graphs $\mathcal{G}$ that consist of multiple strongly connected subgraphs $\mathcal{G}_i$, i.e., $\mathcal{G}_i(E) \subset \mathcal{G}(E)$ and $\mathcal{G}_i(V) \subset \mathcal{G}(V)$. As concluded in an observation in Section 3.1.2, the model reaches a consensus if and only if a globally reachable node exists. Thus, no consensus is reached for these graphs $\mathcal{G}$, as such a node cannot exist by definition.

However, according to [6], the eigenvalue λ_1 is semisimple with the geometric multiplicity m_g and $|\lambda_j| < 1$ for all $j \in \{2, \dots, n\}$ if and only if $\mathcal{G}$ has sinks $\mathcal{S}_i$ (see Section 2) with $i \in \underline{m_g}$, that are all aperiodic. Such graphs $\mathcal{G}$, that fulfill these conditions, therefore, have semi-convergent adjacency matrices $\boldsymbol{A}$, see Theorem 3.1. Moreover, these graphs reach disagreement, instead of consensus, meaning that at least one pair of final opinion values $v_i^{fin} \neq v_j^{fin}$. When calculating the final opinion values, as in [6], we need to differentiate between strongly connected subgraphs and sinks. As the sinks $\mathcal{S}_i$ have no outgoing edges, the final opinion values of the individuals in the sinks are not dependent on other individuals outside of their sink. Therefore, the final opinion value for the nodes $j \in \mathcal{S}_i$, with $i \in \underline{m_g}$, can be calculated as

$$v_j^{fin} = \boldsymbol{r_i p_0} \ ,$$

where $\boldsymbol{r_i}$ is the corresponding left eigenvector, as in Eq. (8). However, the final opinion values of nodes in strongly connected subgraphs that are not sinks, i.e., that have outgoing edges to other strongly connected subgraphs, are dependent on nodes outside of their subgraph. Therefore, the final opinion value for nodes $j \notin \mathcal{S}_i$, with $i \in \underline{m_g}$, is equal to

$$v_j^{fin} = \sum_{p=1}^{m_g} z_{ji}(\boldsymbol{r_i p_0}) \ ,$$

where $\boldsymbol{r_i}$ is the corresponding left eigenvector of the sink to which j is connected. The z_{ji}, with $i \in \underline{m_g}$, are convex combination coefficients that are $z_{ji} > 0$ if and only if there exists a directed path from node j to sink i.

Second, we will discuss how the connectedness of the graph can influence the convergence rate of the model. The conductance Φ of a graph $\mathcal{G}$ is a measure to determine how connected a graph is. Similarly to [9], the measure can be defined for a set S of nodes as

$$\Phi(S) = \frac{\sum_{i \in S, j \notin S} a_{ij}}{min(|S|, |V(\mathcal{G}) \backslash S|)} \ .$$

The numerator $\sum_{i \in S, j \notin S} a_{ij}$ is the sum of the weights of the edges from S to $V(\mathcal{G}) \backslash S$, i.e., it describes how much combined weight the nodes in S give to those in $V(\mathcal{G}) \backslash S$. Therefore, the conductance of a set of nodes S measures the connectedness of S and $V(\mathcal{G}) \backslash S$ compared to how connected the sets are themselves. For a graph $\mathcal{G}$, its conductance is defined as $\Phi(\mathcal{G}) = \min_{S \subset V(\mathcal{G})} \Phi(S)$. Thus, if the conductance of a graph $\mathcal{G}$ is low, there exists a partition of the nodes into two sets (S and $V(\mathcal{G}) \backslash S$), such that the nodes $i, j \in S$ or $i, j \notin S$ are highly connected, while nodes $i \in S$, $j \notin S$ are only sparsely connected. However, if the conductance of the graph $\mathcal{G}$ is large, then the entirety of $\mathcal{G}$ is highly connected. As shown in [9], this measure of conductance can be used to find the upper and lower bounds

$$\frac{\Phi(\mathcal{G})^2}{8} \leq 1 - \lambda_{max} \leq \Phi(\mathcal{G}) \ ,$$

for the eigenvalue λ_{max}. These bounds imply that a higher conductance, i.e., connectedness, of the graph results in a larger spectral gap and therefore in a smaller eigenvalue λ_{max}. Because we observed that λ_{max} is an upper bound for the convergence rate of the model (see Section 3.1.4), we can conclude that highly connected graphs tend to converge faster than sparsely connected ones. This correlation can be observed in Fig. 2 and Fig. 3.

4 Extensions of the DeGroot Model

Although the DeGroot model manages to describe the formation of opinions in a group, as shown in Section 3, it consists of a very simple setup. The only mechanism used to simulate the formation of opinions is the revision of the opinion value vector $\boldsymbol{p_0}$, as shown in Eq. (2). This mechanism merely takes the initial weights into account that the individuals assign to one another at the beginning of the process.

As, however, the formation of opinions in real social groups is far more complex than this iterative process, we will look at two extensions which try to simulate further features of social interaction.

4.1 Friedkin-Johnsen Model

The Friedkin-Johnsen model, as introduced in [3] and later expanded upon in [10], aims to integrate the phenomenon of interpersonal influences into the DeGroot model. This phenomenon describes the social influence of groups on their members through the encouragement of conformity and the discouragement of nonconformity. We will introduce this model, as done in [10].

The basic idea of the Friedkin-Johnsen model is to simulate the impact of these interpersonal influences by using *susceptibility* as an additional metric. The *susceptibility* of an individual i, denoted as d_i, describes how likely this individual is to change their opinion. For example, a non-susceptible individual i would have $d_i = 0$, while a fully susceptible individual j would have $d_i = 1$. To use this metric of *susceptibility* in the model, the matrix

$$\boldsymbol{D} = \begin{bmatrix} d_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & d_n \end{bmatrix} ,$$

is introduced. The interpersonal influences themselves are then expressed through the matrix $\boldsymbol{A}$, used in the DeGroot model, see Section 3.

Like the DeGroot model, the Friedkin-Johnsen model uses an iterative process to revise the opinion values of the individuals in the group. The revised opinion values for the k-th revision are calculated as follows,

$$\boldsymbol{p_k} = \boldsymbol{DAp_{k-1}} + (\boldsymbol{I} - \boldsymbol{D})\boldsymbol{p_0} . \tag{9}$$

If Eq. (9) is used iteratively to repeat the revisions, we obtain

$$\boldsymbol{p_k} = \boldsymbol{V_{k-1}}\boldsymbol{p_0} , \tag{10}$$

where

$$\boldsymbol{V_{k-1}} = (\boldsymbol{DA})^{k-1} + [\sum_{l=0}^{k-2} (\boldsymbol{DA})^l](\boldsymbol{I} - \boldsymbol{D}) . \tag{11}$$

If the model converges, i.e., $\lim_{k\to\infty} \boldsymbol{p_k} = \boldsymbol{p^{fin}}$ exists, then similar to Eq. (3)

$$\begin{aligned} \boldsymbol{p^{fin}} &= \boldsymbol{DAp^{fin}} + (\boldsymbol{I} - \boldsymbol{D})\boldsymbol{p_0} \\ &\iff (\boldsymbol{I} - \boldsymbol{DA})\boldsymbol{p^{fin}} = (\boldsymbol{I} - \boldsymbol{D})\boldsymbol{p_0} , \end{aligned} \tag{12}$$

because when $\boldsymbol{p_k}$ converges asymptotically, then $\boldsymbol{p_k} = \boldsymbol{p_{k-1}}$. Therefore, if $(\boldsymbol{I} - \boldsymbol{DA})$ is invertible, then

$$\boldsymbol{p^{fin}} = (\boldsymbol{I} - \boldsymbol{DA})^{-1}(\boldsymbol{I} - \boldsymbol{D})\boldsymbol{p_0} . \tag{13}$$

Let us assume that at least one $d_i < 1$, otherwise $\boldsymbol{D} = \boldsymbol{I}$, and thus Eq. (9) is $\boldsymbol{p_k} = \boldsymbol{Ap_{k-1}}$, which is precisely Eq. (2), i.e., the model is identical to the DeGroot model. Under this assumption, $(\boldsymbol{DA})$ is row-substochastic, see Section 2. According to [6], if a row-substochastic matrix $\boldsymbol{M}$ is irreducible (see Definition 3.1), then it is also convergent, i.e., $\lim_{k\to\infty} \boldsymbol{M}^k = \boldsymbol{0_{n\times n}}$. In other words, $\boldsymbol{M}$ being irreducible is a sufficient condition for $\boldsymbol{M}$ to be convergent.

Using this theorem, we now provide an independent proof to show when $(\boldsymbol{I} - \boldsymbol{DA})$ is invertible. For this, let us define the following series

$$\boldsymbol{S_n} = \sum_{k=0}^{n} (\boldsymbol{DA})^k = \boldsymbol{I} + \boldsymbol{DA} + \dots + (\boldsymbol{DA})^n .$$

We can rearrange $(\boldsymbol{I} - \boldsymbol{DA})\boldsymbol{S_n} = (\boldsymbol{S_n} - \boldsymbol{DAS_n}) = (\boldsymbol{I} - (\boldsymbol{DA})^{n+1})$. If we assume that $(\boldsymbol{DA})$ is irreducible and, therefore, according to the theorem, convergent, we know that $\lim_{n\to\infty} \boldsymbol{S_n} = \boldsymbol{S^{fin}}$ exists. Thus, $\lim_{n\to\infty}(\boldsymbol{I} - \boldsymbol{DA})\boldsymbol{S_n} = (\boldsymbol{I} - \boldsymbol{DA})\boldsymbol{S^{fin}}$ is well-defined and $(\boldsymbol{I} - \boldsymbol{DA})\boldsymbol{S^{fin}} = \boldsymbol{I}$, as $(\boldsymbol{DA})$ is convergent with $\lim_{n\to\infty}(\boldsymbol{DA})^{n+1} = \boldsymbol{0_{n\times n}}$. Therefore, we can conclude that $(\boldsymbol{I} - \boldsymbol{DA})$ is invertible if $(\boldsymbol{DA})$ is irreducible. Thus, $\boldsymbol{p^{fin}}$ can be calculated with Eq. (13), when $(\boldsymbol{DA})$ is irreducible.

In case $(\boldsymbol{I} - \boldsymbol{DA})$ is not invertible and $\boldsymbol{p^{fin}}$ exists, we can calculate $\boldsymbol{p^{fin}}$ by iteratively applying Eq. (10), until a sufficiently large k is reached [10].

4.2 Hegselmann-Krause Model

People often tend to engage more with others who have similar opinions. Selection and conformation biases are two very well known factors that influence a person's formation of opinions. They both describe how people have a tendency to assign higher weight to information which favors their own opinion. Another effect of this dynamic is that, if an individual i strongly objects to an opinion held by an individual j, it is likely that i will stop interacting with j. The limit for which an opinion can differ for an individual i to still engage with that opinion is referred to as the bounded confidence $\epsilon_i > 0$ of i. To simulate this dynamic, we will look at the Hegselmann-Krause model, as presented in [4]. We will introduce this model, as done in [11] and [12].

Unlike the DeGroot model, the Hegselmann-Krause model assumes that each individual places an equal weight on all the opinion values of the other individuals that hold similar opinions. Thus, a set of neighbors $N_i^k = \{1 \leq j \leq n \mid |v_{ik} - v_{jk}| \leq \epsilon_i\}$ can be defined that contains all individuals with whom i interacts, i.e., those who have a sufficiently similar opinion to i, in revision k. Notice that,

trivially, $i \in N_i^k$ for every k, as $|v_{ik} - v_{ik}| \leq \epsilon_i$ for every ϵ_i. A revision in this model is then calculated as

$$v_{i(k+1)} = \frac{1}{|N_i^k|} \sum_{j \in N_i^k} v_{jk} \,. \tag{14}$$

We can observe that the revised value for a given $v_{i(k+1)}$ is the average opinion value of all neighbors of i in revision k. Moreover, by redefining our adjacency matrix $\boldsymbol{A} \in \mathbb{R}_{n \times n}$ as

$$(\boldsymbol{A}^k)_{ij} = \begin{cases} \frac{1}{|N_i^{k-1}|} & \text{if } j \in N_i^{k-1}, \\ 0 & \text{else.} \end{cases},$$

we can calculate the revision through $\boldsymbol{p_k} = \boldsymbol{A}^k \boldsymbol{p_{k-1}}$, similar to Eq. (2).

Notice that this model describes a DeGroot model, where all outgoing edges from a certain node have the same weight, and edges between two nodes can be added and removed depending on the similarity of their opinion values.

Furthermore, to observe under which conditions the Hegselmann-Krause model reaches asymptotic consensus, we define an ϵ-chain, as done in [11] and [12].

Definition 4.1 (ϵ-chain). *A vector $\boldsymbol{p_k} \in \mathbb{R}_+^n$ is called an ϵ-chain with $\epsilon \geq 0$ if an ordering $v_{1k} \leq \cdots \leq v_{nk}$ exists, such that $v_{(i+1)k} - v_{ik} \leq \epsilon$ for all $i \in \{1, \ldots, n-1\}$.*

With this definition, we can show in the following that an instance of the Hegselmann-Krause model, with $\epsilon_i = \epsilon$ for all $i \in \underline{n}$, reaches asymptotic consensus if and only if the vector $\boldsymbol{p_k}$ is an ϵ-chain for all revisions k [11]. For this, we offer a short proof based on previous observations.

Proof. Let us assume that $\boldsymbol{p_k}$ is an ϵ-chain for all revisions k. As we observed for the DeGroot model (in Section 3.1.2), the model necessarily reaches a consensus if a globally reachable node exists. As $\boldsymbol{p_k}$ is an ϵ-chain for all $k \in \mathbb{N}$, the corresponding graph $\mathcal{G}$ is, by definition, strongly connected in each revision. This trivially satisfies the necessary condition for convergence. In the other direction, if therefore $\boldsymbol{p_k}$ is not an ϵ-chain for at least one revision k, then the ordering of the opinion values must have at least two neighboring individuals that are not connected in that revision. Thus, there exists no globally reachable node and the model does not converge. □

Moreover, for these instances that reach consensus, we can use $v_k^{max} = v_{jk}$ with $v_j = \max_{i \in \underline{n}} v_i$ and $v_k^{min} = v_{j'k}$ with $v_{j'} = \min_{i \in \underline{n}} v_i$, i.e., the opinion values in revision k of the individuals with the highest and lowest initial opinion values, as an upper and lower bound for all the other opinion values. They can therefore be used as upper bounds for the convergence rate, similar to the upper bound in Section 3.1.4. For this, we offer an independent proof.

Proof. Let us assume again that $\boldsymbol{p_k}$ is an ϵ-chain for all revisions k. We will prove by induction that the values v_k^{max} and v_k^{min} are upper and lower bounds with $v_k^{min} \leq v_{ik} \leq v_k^{max}$ for all $i \in \underline{n}$ in every revision k. The statement is trivially true for $k = 0$, as v_0^{max} and v_0^{min} are precisely defined as the largest and smallest of the initial opinion values. Assume that the statement is correct for any $k \in \mathbb{N}$. For any revision $k+1$, let us assume that there exists $j \in \underline{n}$ such that $v_{j(k+1)} > v_{k+1}^{max}$. Notice that if $i \in N_{max}^k$ and $l \in N_j^k$ with $v_{lk} \geq v_{ik}$, then $l \in N_{max}^k$, as l must be in the bounded confidence $\epsilon_{max} = \epsilon$ around v_k^{max}. Thus, $avg_{i \in N_{max}^k}(v_{ik}) \geq avg_{i \in N_j^k}(v_{ik})$, i.e., the average opinion value of the neighbors of v_k^{max} is greater than the average opinion value of the neighbors of v_{jk}. As we observed for Eq. (14), the revised values v_{k+1}^{max} and $v_{j(k+1)}$ are the averages of the opinion values of the neighbors of max and j in revision k. Therefore, $v_{k+1}^{max} \geq v_{j(k+1)}$. The same argument can be used to show that v_k^{min} is a lower bound. □

An example of the model can be seen in Fig. 4. Although the nodes with the initial values 4 and 6 do influence each other in the first iteration, i.e., they have a connecting edge, they are then pulled in opposite directions. Thereby, they no longer communicate, as they are outside of each other's bounded confidence, and converge towards different opinion values, as the ϵ-chain is broken.

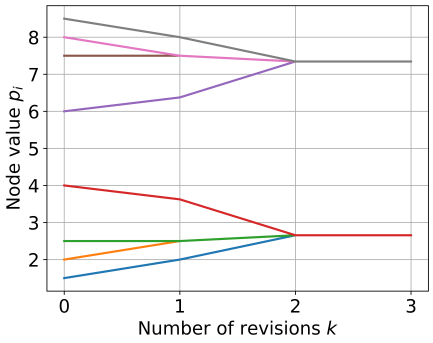


Fig. 4: The figure shows an instance of the Hegselmann-Krause model with the bounded confidence value $\epsilon = 2$.

5 Conclusion and Outlook

In this paper, we examined the DeGroot model to analyze the formation of opinions in groups. We then reproduced in Theorem 3.1 under which conditions the model converges, as well as a sufficient condition (primitive matrices) and a necessary condition (existence of a globally reachable node) for a convergence towards a consensus. The paper also shows that only the left eigenvector $\boldsymbol{r^T}$ and the vector of initial opinion values $\boldsymbol{p_0}$ are necessary to compute the consensus value. Moreover, we showed that the second largest eigenvector in magnitude λ_{max} is a good measure for the convergence rate of the model. Furthermore, we looked at how the disagreement of graphs with multiple strongly connected subgraphs can be calculated and how the connectedness of a graph can be used as a further measure for the convergence rate. Lastly, we expanded on the DeGroot model by examining the additional features: *susceptibility* and *confidence bounds*.

In the future, further extensions that simulate other phenomena could be taken into account. For example, repulsive behavior ("negative weights") that allows for a change in the opposite direction of the other individuals opinion values [13], interrelated topics, as a subject rarely exists in isolation [14], or a distinction between publicly versus privately expressed opinions (especially opinions on highly stigmatized topics, are often expressed only to certain individuals) [15].

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